

TrES-3b

R-2684

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_220709 · created 2026-08-31T00:26:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459769.8592 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.055
Transit depth (Rp/Rs)^2	2.52 +/- 1.75 [%]
Orbital Inclination (inc)	78.82 +/- 2.94
Airmass coefficient 1 (a1)	39.17 +/- 23.46
Airmass coefficient 2 (a2)	-2.62 +/- 0.33
Scatter in the residuals of the lightcurve fit is	117.95 %
Best Comparison Star	#1 - [392, 418]
Optimal Aperture	1.9
Optimal Annulus	8.87
Transit Duration (day)	0.021 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.159 $\pm$ 0.055 vs 0.16309 (deviation 0.04 σ)
Duration fit vs published	0.021 d vs 0.0567 d (deviation 0.74 σ)
Mid-time O-C	1.6 min
Depth SNR	1.6
Residual scatter	117.95 %

Light curve

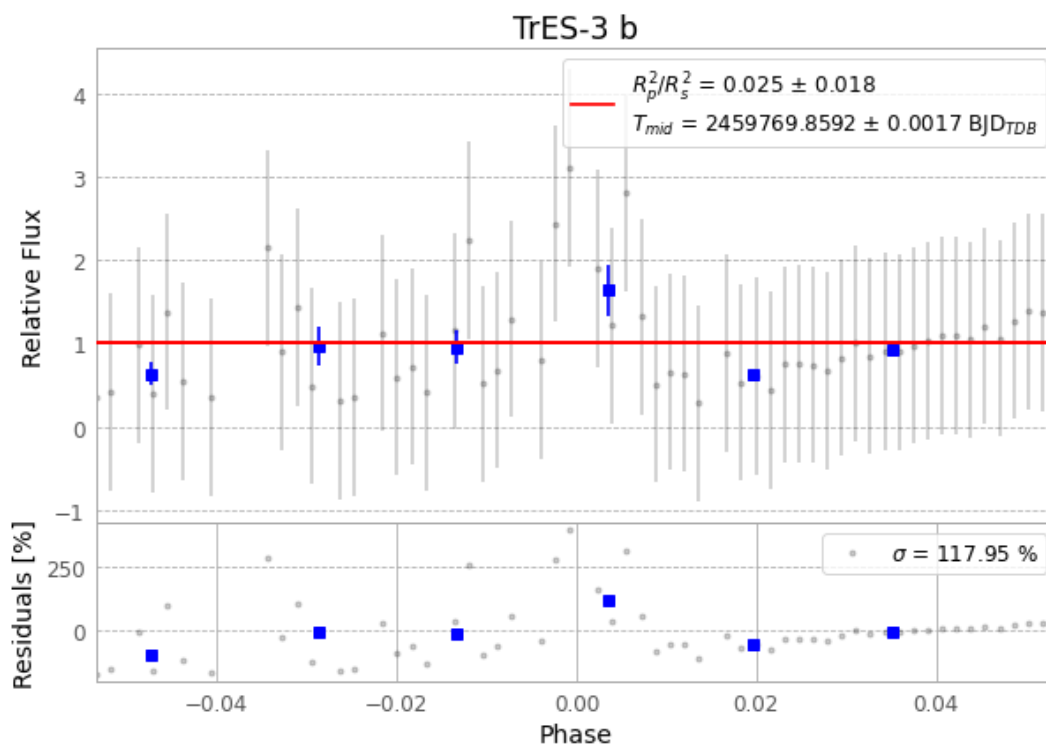


Figure 1 — FinalLightCurve_TrES-3 b_2022-07-08.png (SHA-256 eb92dd26e79bfb64...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 328], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c82982c8bae656a2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

68 FITS frames (45 MB), TRES-3220709065113.FITS → TRES-3220709101213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-220709.log (c7f6ab5374ee...), FinalLightCurve_TrES-3 b_2022-07-08.png (eb92dd26e79b...), FinalLightCurve_TrES-3 b_2022-07-08.csv (831e53e6e60b...)

Compute resources

Wall time 287.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 00:26Z.